

Assignment - 1 CSCU

Section - A

Q.1 Phishing is a cyberattack in which attackers impersonate a trusted entity to trick individual into revealing sensitive information such as passwords or bank details.

Two common techniques :-

- Email Phishing :- Fake e-mail that appear from legitimate organizations asking users to click malicious links.
- Spear Phishing :- Targeted phishing attacks aimed at a specific individual or organization using personalized information.

Q.2 Malware is malicious software designed to damage, disrupt or gain unauthorized access to a computer system.

E.g :- (i) Virus : Attaches to legitimate files and spreads when the file is executed.

(ii) Ransomware : Encrypts user data and demands payment to restore access.

Q.3 Encryption protects data by converting it into unreadable form, ensuring confidentiality and preventing unauthorized access.

Examples of encryption algorithms :-

- AES (Advanced Encryption Standard)
- RSA (Rivest - Shamir - Adleman)

Q.4 • Symmetric Encryption :- Uses the same key for encryption and decryption.
E.g :- AES.

- Asymmetric Encryption :- Uses a public key for encryption and a private key for decryption.
E.g :- RSA.

Q.5 SQL Injection is a cyberattack where malicious SQL queries are inserted into input fields to manipulate a database.

Two consequences :-

- Unauthorized ~~acc~~ access to sensitive data.
- Deletion or modification of database records.

Section - B

Q.1 Social engineering is a technique where attackers manipulate people into revealing confidential information.

E.g :-

1. Phishing :- Fake emails to steal login details.

Mitigation :- User awareness training and email filtering.

2. Pretexting :- Creating a fake scenario to obtain information.

3. Baiting :- Offering something attractive (e.g., free USB) to infect systems.

Mitigation :- Avoid unknown devices and implement security policies.

Q.2 A firewall is a security device that monitors and controls incoming and outgoing network traffic based on security rules.

Types :

1. Packet-Filtering Firewall :- Filters traffic based on IP address, port number, and protocol.

2. Application-Level Firewall :- Inspect data at the application layer for deeper analysis.

Comparison :- Packet-filtering is faster but less secure, while application level firewalls provide better security but may reduce performance.

Q.3 A Distributed Denial of Service (DDoS) attack floods a network or server with excessive traffic from multiple sources, making it unavailable to user.

Mitigation methods :

1. Traffic Filtering and Rate Limiting.
2. Use of Content Delivery Networks (CDNs).
3. DDoS Protection Services and Firewalls.

Section - C

Q.1 A Vulnerability assessment identifies security weakness in systems and networks.

Steps :

1. Asset identification
2. Threat identification
3. Vulnerability scanning
4. Risk assessment
5. Reporting and remediation.

Vulnerability scanning tools automatically detect known security flaws, misconfigurations and outdated software.

E.g :-

- Nessus
- Open VAS

These tools improve efficiency and accuracy in identifying risks.

Q.2 Penetration testing is a simulated cyberattack conducted to evaluate system security.

White - Box Testing : Tester has full knowledge of system details.

- Advantages : Thorough and detailed assessment.
- Disadvantages : May not simulate real attacker behavior fully.

Black - Box Testing : Tester has no prior knowledge of the system.

- Advantages: Simulates real-world attacks.
- Disadvantages: May miss internal vulnerabilities.

Q.3 Incident Response is a structured approach to handling cybersecurity incidents.

Phases :

1. Preparation: Develop policies, tools and training.
2. Identification: Detect and confirm the incident.
3. Containment: Limit the spread of the attack.
4. Eradication: Remove the root cause (malware, vulnerabilities).
5. Recovery: Restore systems to normal operation.
6. Lessons Learned: Analyze the incident and improve defenses.

Importance : An effective incident response plan minimizes damages, reduces recovery time, ensure business continuity, and protects organizational reputation.